

Deciphering the critical degradation factors of solid composite electrodes with halide electrolytes: Interfacial reaction versus ionic transport

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ABSTRACT

Recently, halide-type Li^+ conductors have been revisited for their use in all-solid-state batteries (ASSBs) owing to their stability at high potentials. However, the realization of ASSBs is hindered by the fast performance decay of composite cathodes. From a comparative study using halide and sulfide solid electrolytes (SEs), herein, we reveal the critical degradation factors of halide-SE-based cathodes, which are different from the conventional findings of sulfide-SE-based cathodes. By using impedance decoupling combined with scanning spreading resistance microscopy and force spectroscopy, we elucidate the mechanisms behind the SE-dependent degradation of single-particle $\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$ (NCM) composite cathodes. Impedance analyses show that NCM- $\text{Li}_6\text{PS}_5\text{Cl}$ (LPSCl) and NCM- Li_3InCl_6 (LIC) exhibit considerable increase in interfacial impedance and Li^+ -transport impedance, respectively, upon cycling. Based on the combined experimental and computational study of microscopic interfacial and mechanical properties, we demonstrate that the degradation of NCM-LPSCl originates primarily from the formation of resistive interphases, while the crucial degradation factor of NCM-LIC is the cracking-induced mechanical deformation of the LIC under pressure. Finite element analysis results further reveal how the deformation behavior of the SE materials influences the formation and propagation of cracks in composite cathodes during cycling. This study provides insights into the design of materials and electrodes for ASSBs with high power capabilities and long cycle lifetimes.

1. Introduction

Li-ion batteries (LIBs) have been widely employed in industries, such as electronic devices, electric vehicles, and energy storage systems because of their high power density and long life cycle despite their compact size and light weight [1,2]. However, as flammable organic liquid electrolytes in commercial LIBs raise considerable safety concerns [3,4], all-solid-state batteries (ASSBs) with inorganic solid electrolytes have attracted increasing research interest [5–8]. Further, several key advantages of ASSBs compared to LIBs, including their wide operating temperature range and ability to increase their energy density, have also gained considerable attention [9–12]. Unfortunately, the unsatisfactory electrochemical performance of ASSBs, originating from the rapid

degradation of composite cathodes comprising cathode materials, SEs, and conductive agents, still remains a challenge for their practical utilization [13].

Recently, ASSBs based on sulfide SEs have been extensively studied because of their high ionic conductivities. Particularly, argyrodite-type ($\text{Li}_6\text{PS}_5\text{X}$, X = Cl, Br, and I) sulfide SEs and their derivatives have achieved excellent room-temperature ionic conductivities that are comparable to those of liquid electrolytes [14–17]. Further, their mechanical sinterability enables facile cell fabrication using a simple cold pressing process [18]. However, highly reactive sulfide SEs are chemically unstable in air and moisture, thereby resulting in the decomposition of the sulfide phase, which considerably deteriorates the ionic conductivity and releases toxic H_2S gas [19–21]. Furthermore, the intrinsically low

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electrochemical oxidation stability of sulfide SEs resulted in oxidative side reactions with high-voltage cathode materials.

Halide SEs have emerged as attractive alternatives to address the stability problem of sulfide SEs owing to their high electrochemical oxidation stability of halogen ions compared to that of a sulfur ion. Ternary halides with monoclinic crystal structures (Li_3MX_6 , $\text{M} = \text{Y, Er, In}$; $\text{X} = \text{Cl, Br, I}$) have demonstrated high electrochemical oxidation stability ($> 4 \text{ V vs. Li/Li}^+$), good chemical compatibility with cathode materials, and improved air stability [22–25]. Although halide SEs can suppress oxidative side reactions in composite cathodes, the performance degradation of ASSBs remains a challenge, as reported in previous studies [26–30].

In this study, based on the mechanistic study of SE-dependent degradation of composite cathodes using sulfide- and halide-SE, we revealed the critical degradation factors of halide-SE-based cathodes. To avoid any complications from the microstructural degradation of the active materials, single-particle $\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$ (NCM) with enhanced mechanical robustness was employed [31–35]. Particularly, we compared the degradation behavior of composite cathodes by decoupling the impedance signals using a modified transmission line model (TLM) [36,37]. The impedance analysis showed that the degradation of NCM- Li_3InCl_6 (LIC) and NCM- $\text{Li}_6\text{PS}_5\text{Cl}$ (LPSCl) was accompanied by increased ionic transport and interfacial charge transfer impedances, respectively. To understand the distinctive degradation behaviors of halide-SE-based cathodes, we investigated the electrical and mechanical properties of composite cathodes using scanning spreading resistance microscopy (SSRM) and force spectroscopy (FS). The SSRM and FS studies revealed that the major degradation factor of NCM-LIC was the cracking-induced mechanical deformation of the LIC under pressure, while the degradation of NCM-LPSCl was driven by the formation of resistive interphases. Furthermore, a computational finite element analysis was conducted to investigate the mechanism of crack formation and propagation.

2. Experimental

2.1. Materials

Single-particle NCM powder was obtained from COSMO AM&T (Republic of Korea). LPSCl powder was mechanochemically synthesized using lithium sulfide (Li_2S , 99%, Sigma-Aldrich), phosphorus(V) sulfide (P_2S_5 , 99%, Sigma-Aldrich), and lithium chloride (LiCl , 99.9%, Sigma-Aldrich) by high-energy ball milling at 600 rpm for 10 h with ZrO_2 balls using Pulverisette 7 PL (Fritsch GmbH). Similarly, LIC powder was prepared by high-energy ball milling of a stoichiometric mixture comprising LiCl and indium chloride (InCl_3 , Sigma Aldrich, 99.9%) at 600 rpm for 3 h. Further, the resultant LIC powder was annealed at 260°C under a vacuum for 12 h.

2.2. Material characterizations

The microstructures, morphologies, and compositions of the materials were investigated using scanning electron microscopy (SEM, Hitachi S-4800), coupled with energy dispersive X-ray spectroscopy (EDS). To observe the morphology of the composite cathodes, the cells were disassembled and the composite cathodes were polished to remove the remaining SEs. To obtain cross-sectional backscattered electron-SEM images, the cathodes were collected after charging to 4.3 V vs. Li/Li^+ in an Ar-filled glove box and polished by an Ar ion beam using a cross-sectional polisher (Hitachi ArBlade5000). The structural characterization of the NCM, LPSCl, and LIC powders was performed by powder X-ray diffraction (XRD) measurements with $\text{Cu K}\alpha$ radiation ($\lambda = 1.541 \text{ \AA}$) using an X-ray diffractometer (Horiba, Miniflex 600 Mini) in a dry room. X-ray photoelectron spectroscopy (XPS) spectra were acquired using a monochromatic $\text{Al K}\alpha$ source (1486.6 eV) in an ESCALAB 250Xi spectrometer (Thermo Scientific). To observe the local resistivity

distribution of the composite electrodes, the cross-sectional surfaces of the composite cathodes were investigated using atomic force microscopy (AFM) with SSRM (Park Systems, Co., Ltd., NX-10 with a logarithm amplifier module) with a +2.5 V DC bias from a conductive platinum-coated tip (CONTSCPT) in a dry room. During SSRM measurements, a DC bias is applied between the specimen and probe tip. Then, electrons flow from the tip to the conductive part of the specimen, and the corresponding resistance is recorded. 3D resistivity mapping images ($2 \mu\text{m} \times 2 \mu\text{m}$) were constructed using z-height data from the topography scan and resistivity at the cross-sectional surface. To acquire modulus mapping results of SEs, we utilized the scanning FS mode of AFM with a silicon probe (PPP-NCHR). During FS measurements, the cantilever approaches and retracts repeatedly on each scanning site (256×256 pixels), recording the relation between z-height dislocation and force that is imposed to the cantilever. For both SSRM and FS, a scan rate was fixed to 0.2 Hz. Detailed information about the SSRM and FS scanning is presented in Supplementary Note 1 (Fig. S1) and 2, respectively.

2.3. Electrochemical experiments

Composite cathodes were prepared by mixing single-particle NCM, SE, and Super P in a weight ratio of 72:27:1 using a ball mill with ZrO_2 balls at a speed of 250 rpm for 48 h. All-solid-state cells with a diameter of 10 mm, comprising stainless-steel rods as the current collectors and a polyetheretherketone (PEEK) mold, were assembled using the following procedure. After pelletizing the SE layer with 100 mg of LPSCl powders at 100 MPa, the as-prepared powder mixture was pressed on one side at 433 MPa. Additionally, indium (100 μm , 10 mm in diameter) and lithium foils (200 μm , 4 mm in diameter) were placed on the other side of the SE layer. Further, after inserting a stainless-steel foil (10 μm thick, current collector) on both sides, the cell was pressed at 250 MPa.

The cells were subjected to galvanostatic cycling in the voltage range of 2.5 to 4.3 V vs. Li/Li^+ at 30°C . The formation step was performed at 0.05C for two cycles, where the charge step was conducted in a constant current-constant voltage (CC–CV) mode with CC charging to 4.3 V vs. Li/Li^+ , followed by CV charging with a cut-off current of 0.01C. After the formation step, charge-discharge cycling was conducted with CC charging at 0.33C, followed by CV charging with a cut-off condition of 0.06C. The discharge rate was 0.33C. The rate capability was examined at C-rates of 0.1, 0.2, 0.5, 1, and 2C. To analyze the electronic, ionic, and interfacial resistance of a composite electrode, two types of cells, ion-blocking [stainless-steel | composite electrode | stainless-steel] and full [stainless-steel | composite electrode | LPSCl | Li-In alloy | stainless-steel] cells were prepared and impedance spectra of each cell were investigated by a potentiostat (Biologic SP-300) at 30°C . The measurements were performed in the frequency range of 7 MHz to 1 mHz with an AC amplitude of 5 mV.

2.4. Computational method

Finite element analysis was used to investigate volume change-induced stress and crack propagation in SEs. Purely mechanical deformation caused by the delithiation and lithiation of NCM was considered. The size of the NCM particles was fixed at 3 μm . The grain structure of the SE was generated using the Voronoi tessellation method, with an effective size of 10 μm , as shown in Fig. S2. To match the volume fractions of NCM and SEs, the domain size was set to be 12.55 μm . The mechanical moduli of NCM, LPSCl, and LIC were 175 GPa, 32.8 GPa, and 41.1 GPa, respectively. The volume change of NCM particles was modeled using an analogy to thermal strain. Accordingly, the diffusion-induced strain (ε) of NCM was governed by $\varepsilon = \Omega/3 \times \Delta c$, where ε is the compositional strain caused by Li insertion and extraction, Ω is the partial molar volume calculated based on the 5.9% volume change of NCM ($1.338 \times 10^{-6} \text{ m}^3 \text{ mol}^{-1}$) [38], and c is the Li concentration. The mechanical yield strength of LPSCl was assumed to be 200 GPa that was

similar to that of sulfide SEs, such as $\text{Li}_2\text{S-P}_2\text{S}_5$ [39]. The critical energy release rate for crack tip opening was parameterized within a range of 0.5 to 3 J m^{-2} , since sulfide SEs are known to have a value of 2.93 J m^{-2} [40]. The interfacial strength of grain boundaries was assumed to be 30% weaker than that within the grain. For crack propagation inside SE grains, finer Voronoi structures were discretized to a few hundred nanometers. Cohesive zone elements were assumed on the edges of the second generated Voronoi tessellation (Fig. S2). Details of the crack propagation model are described in the Supplementary Note 3.

3. Results and discussion

To gain an in-depth understanding of the critical degradation factors, two different NCM composite electrodes with LIC and LPSCI SEs were prepared, and their degradation behaviors were compared (Fig. 1a). Both SEs were synthesized using a mechanochemical process, and their particle sizes were determined to be approximately $5 \mu\text{m}$ using SEM (Fig. S3). From AC-impedance measurements, the Li^+ conductivities of

LIC and LPSCI were measured as 1.2 and 1.6 mS cm^{-1} , respectively, at 25°C (Fig. S4). XRD patterns of LIC and LPSCI corresponded with those of monoclinic (ICSD No. 418,490) and cubic (ICSD No. 04-009-9027) phases without any impurity phases (Fig. S5) [41,42]. Single-particle NCM (sizes $\sim 1\text{--}5 \mu\text{m}$) with a smooth surface morphology was employed as the cathode active material instead of conventional, multi-particle NCM comprising nanosized agglomerates (Fig. S6). Using mechanically robust single-particle NCM enabled us to avoid any complications arising from the cracking and pulverization of NCM particles during cycling, and thus clearly observe the degradation phenomena governed by the SE [43].

The composite cathodes were combined with an SE layer of LPSCI (thickness $\sim 200 \mu\text{m}$) and an Li-In alloy anode to construct ASSBs. As shown in Fig. S6, NCM-LPSCI exhibited slightly lower cell voltages at the beginning of the first charging (formation) process, which is known to have originated from the interfacial side reaction between NCM and LPSCI [44,45]. Contrarily, there was no sagging part in the voltage profile of NCM-LIC, thereby revealing the electrochemical stability of

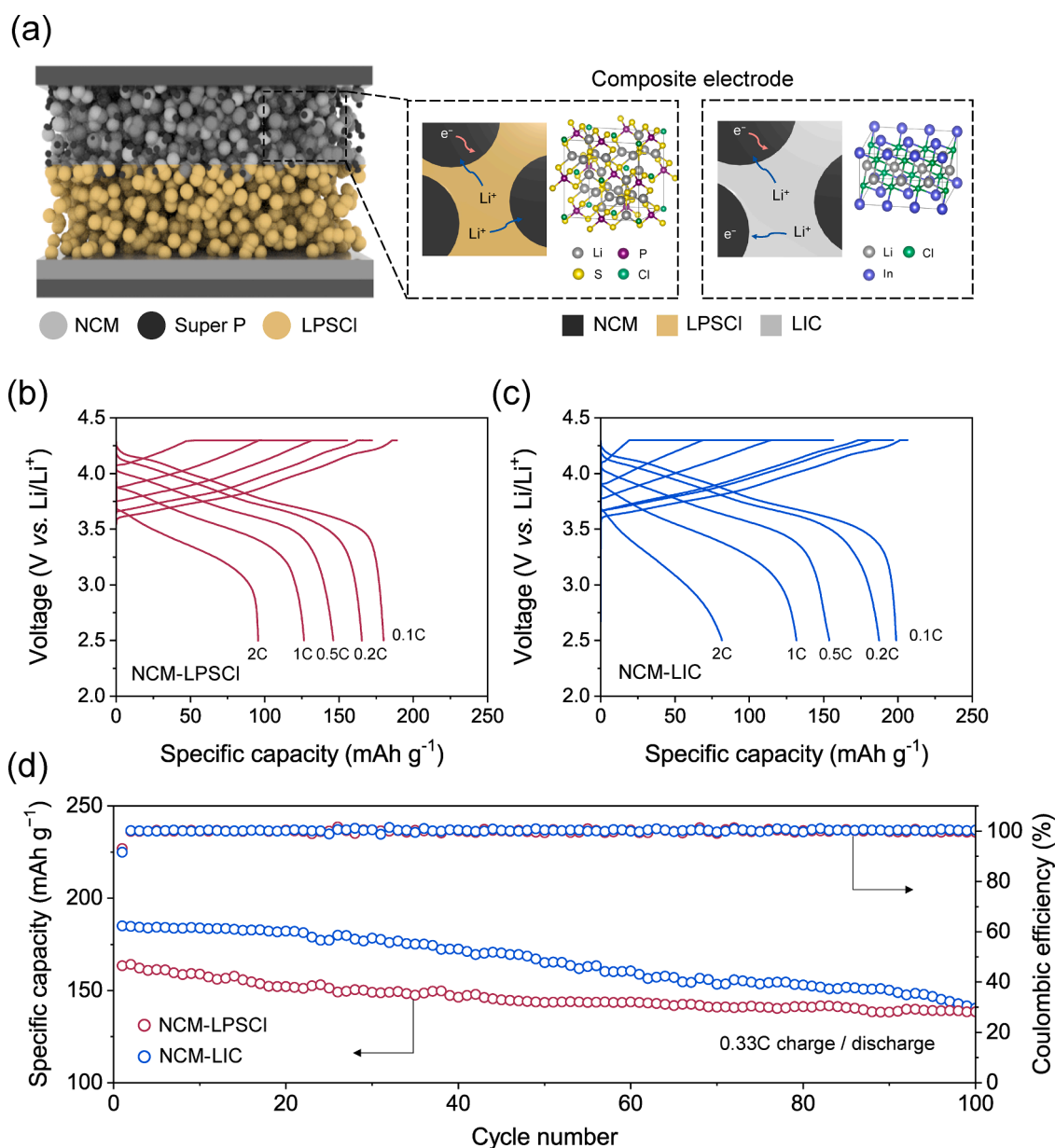


Fig. 1. (a) Schematic of an ASSB with NCM-LPSCI and NCM-LIC. Voltage profiles of ASSBs with (b) NCM-LPSCI and (c) NCM-LIC cathodes during the rate-capability test from 0.1C to 2.0C. (d) Capacity retentions and Coulombic efficiencies of ASSBs with NCM-LPSCI and NCM-LIC.

the NCM/LIC interface. Consequently, NCM-LIC showed a higher discharge capacity (197.7 mAh g^{-1}) and initial Coulombic efficiency (89.5%) compared to those of NCM-LPSCI (186.4 mAh g^{-1} and 84.5%). Figs. 1b and c present the voltage profiles of ASSBs using NCM-LIC and NCM-LPSCI, respectively, at different rates from 0.1 to 2C. At low rates of 0.1–1C, NCM-LIC delivered larger discharge capacities than NCM-LPSCI. However, the former showed an inferior performance compared to the latter at 2C, which was characterized by a larger charge-discharge voltage gap and a smaller discharge capacity. The result of rate capability test suggests that NCM-LPSCI and NCM-LIC have different charge transport and interfacial reaction behavior, which can be decoupled by AC impedance spectroscopy. The electrochemical behaviors of ASSBs during long-term cycling tests at 0.33C are compared in Figs. 1d and S7. Despite the higher efficiency and larger initial discharge capacity, the cell with the LIC cathode exhibited a faster performance drop with a reduced capacity of 140.4 mAh g^{-1} (capacity retention = 75.7%) after 100 cycles compared to that of the LPSCI cathode (138.5 mAh g^{-1} , capacity retention = 84.7%) after 100 cycles.

To determine the degradation-determining factors of the composite cathodes upon cycling, AC impedance spectra were collected and analyzed using a modified TLM. Fig. 2a shows a TLM-based equivalent circuit describing multiple processes in a solid composite cathode. The TLM used in this study is expressed by a network of distributed impedance elements which considers Li^+ transport, electronic transport, and interfacial reactions at the NCM/SE interface combined with solid-state diffusion in the NCM [41]. In the circuit, r_{ion} and r_{ele} designate the resistances per unit length ($\Omega \text{ cm}$) for the Li^+ and electronic transport processes, respectively. z_{int} denotes the impedance per unit volume ($\Omega \text{ cm}^3$) and comprises r_{int} , cpe_{int} , and z_{w} , which correspond to the resistance, constant phase element, and Warburg impedance for the interfacial reaction and solid-state diffusion in NCM, respectively. The measured impedance spectra of the pristine and cycled cathodes were

fitted to the equivalent circuit of the TLM to decouple the different impedance elements (Fig. 2b). The resulting parameters are listed in Table S1. The following three points should be mentioned regarding decoupled impedance elements. First, the pristine NCM-LIC cathode showed a considerably lower r_{int} value ($1.3 \Omega \text{ cm}^3$) than that of NCM-LPSCI ($6.8 \Omega \text{ cm}^3$), as expected from the earlier discussion on the first charge curves (Fig. S7). However, it exhibited a higher r_{ion} value ($15,125 \Omega \text{ cm}$) than that of NCM-LPSCI ($6,305 \Omega \text{ cm}$) due to a lower Li^+ conductivity. The rate capability data of NCM-LPSCI and NCM-LIC in Figs. 1b and c can be understood in terms of different dominant factors that determine their charge-discharge performances: interfacial reaction for NCM-LPSCI and ionic transport for NCM-LIC. At low C-rates, the interfacial resistance dominates the cell performance, and thus, NCM-LPSCI with resistive interphases delivers lower capacities, as compared to NCM-LIC. At high C-rates, the kinetics of interfacial charge transfer is enhanced by large driving forces (overpotentials applied across the interface), whereas the IR drop due to the resistance for ionic transport becomes very high. Therefore, NCM-LIC with lower Li^+ conductivity exhibits larger charge-discharge voltage gaps and lower capacities in comparison to NCM-LPSCI. Second, NCM-LIC showed a 2-fold increase in r_{ion} from 15,125 to 29,780 $\Omega \text{ cm}$ after 100 cycles, while r_{int} increased slightly to $2.3 \Omega \text{ cm}^3$. Thirdly, the r_{ion} value of NCM-LPSCI became slightly larger upon cycling. However, r_{int} increased considerably from 6.8 to 19.6 $\Omega \text{ cm}^3$. Thus, it was obvious that the degradation behaviors of both cathodes were governed by different critical factors: Li^+ transport for NCM-LIC versus interfacial reactions for NCM-LPSCI.

To understand the impedance characteristics of NCM-LIC and NCM-LPSCI upon cycling, the electrical properties of the cycled cathodes were investigated using AFM, which was equipped with SSRM (Supplementary Note 1 and Fig. S1). Figs. 3a and b present the two- (2D) and three-dimensional (3D) mapping results of the electronic resistivity acquired on the cross-sections of NCM-LIC and NCM-LPSCI after 100 cycles. SSRM

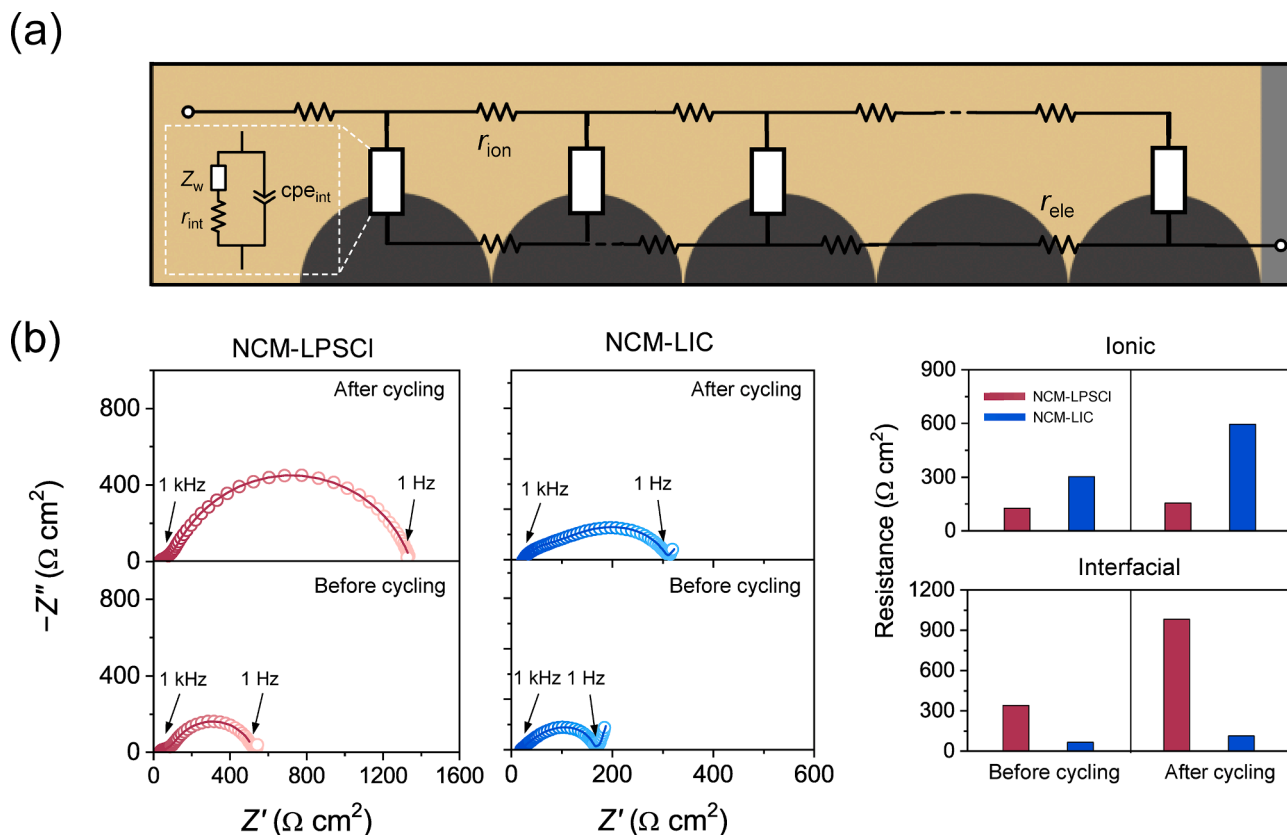


Fig. 2. (a) Equivalent circuit for the TLM with interfacial reactions. (b) AC-impedance spectra of NCM-LPSCI and NCM-LIC with corresponding ionic and interfacial resistances.

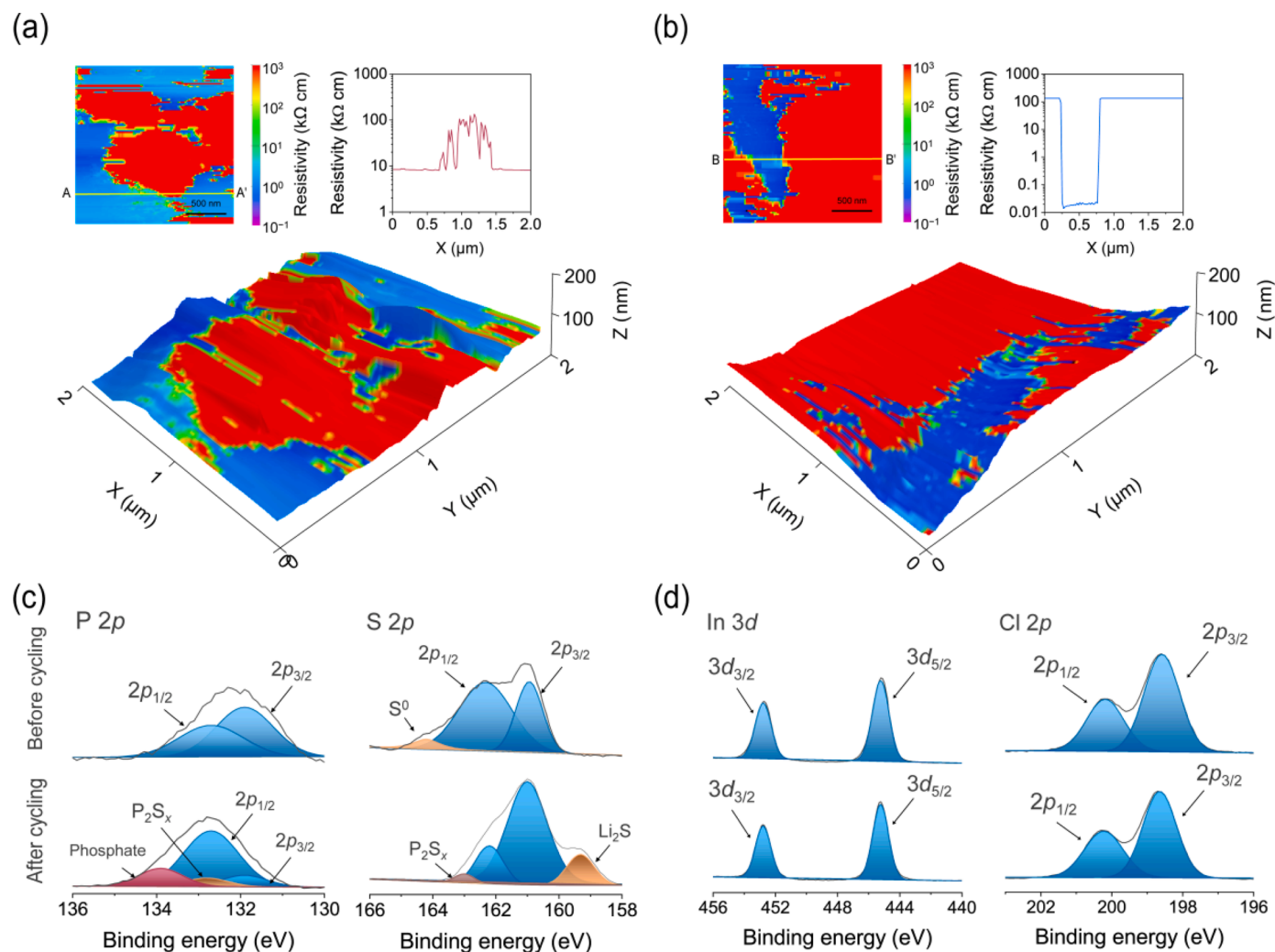


Fig. 3. SSRM mapping results of (a) NCM-LPSCl and (b) NCM-LIC after cycling. The 2D scanning plots (top left in (a) and (b)) shows the resistivity values displayed over an X-Y plane. The yellow lines (A-A' and B-B') indicates the selected regions for the line scan. The line scan of resistivity (top right in (a) and (b)) displays the change in resistivity at the SE-active material interfaces. The 3D resistivity-topology scan (bottom) combines the resistivity data with z-height data, providing a simultaneous visualization of the topology of the sample and the corresponding resistivity. XPS spectra for (c) P 2p, S 2p of NCM-LPSCl and (d) In 3d, Cl 2p of NCM-LIC.

was conducted in a scan area of $2 \mu\text{m} \times 2 \mu\text{m}$. The vertical axis in the 3D plot represents the topology (height variation) of the composite cathode in the region of interest. The SSRM result showed two regions corresponding to the NCM phase with low resistivity (green to blue in color, $< 10 \text{ k}\Omega \text{ cm}$) and the SE phase with extremely high resistivity (red). On the one hand, a high-resolution scanning measurement of the NCM-LPSCl cathode over a line (A-A' in Fig. 3a) revealed a variation in resistivity across the interfacial region ($\sim 50 \text{ nm}$), thereby suggesting the presence of a resistive interphase. On the other hand, the line scan profile (B-B' in Fig. 3b) of NCM-LIC showed discontinuity in resistivity at the NCM/LIC interface, that is, no continuous variation in the resistivity over a finite distance in the interfacial region.

The chemical states of the composite cathodes after cycling were investigated using XPS. Fig. 3c compares the P 2p and S 2p XPS spectra of the pristine and cycled NCM-LPSCl cathodes. The P 2p spectrum exhibited the characteristic peaks of argyrodite at binding energies of 131.9 and 132.7 eV before cycling [46–48]. After cycling, additional peaks for P_2S_x (132.8 eV) and phosphates (133.7 eV) appeared, owing to the oxidation of LPSCl and side reactions of LPSCl with oxygen-containing NCM to form PO_4^{3-} polyanions at the interface [47]. Further, a comparison of the S 2p spectra obtained from pristine and cycled NCM-LIC revealed the formation of sulfide species (Li_2S and P_2S_x)

upon cycling on the one hand [48]. On the other hand, In 3d and Cl 2p spectra of NCM-LIC (Fig. 3d) remained unchanged even after cycling, thereby supporting the electrochemically stable nature of LIC during cycling [49,50]. The resistivity mapping and XPS analyses showed that the degradation of NCM-LPSCl was mainly caused by the formation of highly resistive phosphorus- and sulfur-based interphases. This corresponds with the impedance decoupling results, thereby showing a significant increase in r_{int} for cycled NCM-LPSCl. At this point, it should be emphasized that unlike NCM-LPSCl, the interface between NCM and LIC remained relatively stable during cycling, as confirmed by the combined impedance-SSRM-XPS characterizations. However, a question arose regarding NCM-LIC: “What makes Li^+ to transport sluggishly and causes performance degradation?”

To uncover a clue as to the observed increase in r_{ion} upon cycling (Fig. 2b), we prepared SE-only pellets and monitored the evolution of Li^+ conductivity under the application of a constant pressure of 250 MPa. Generally, Li^+ conductivity increases with the applied pressure on the electrolyte due to the improved solid-solid contact between electrolyte particles. The pressure-induced change in Li^+ conductivity depends on the deformation behavior of the electrolyte. As shown in Fig. 4a, the Li^+ conductivity of LIC exhibited a decreasing trend in time from 1.2 to 0.6 mS cm^{-1} over 240 h. Contrarily, LPSCl retained its

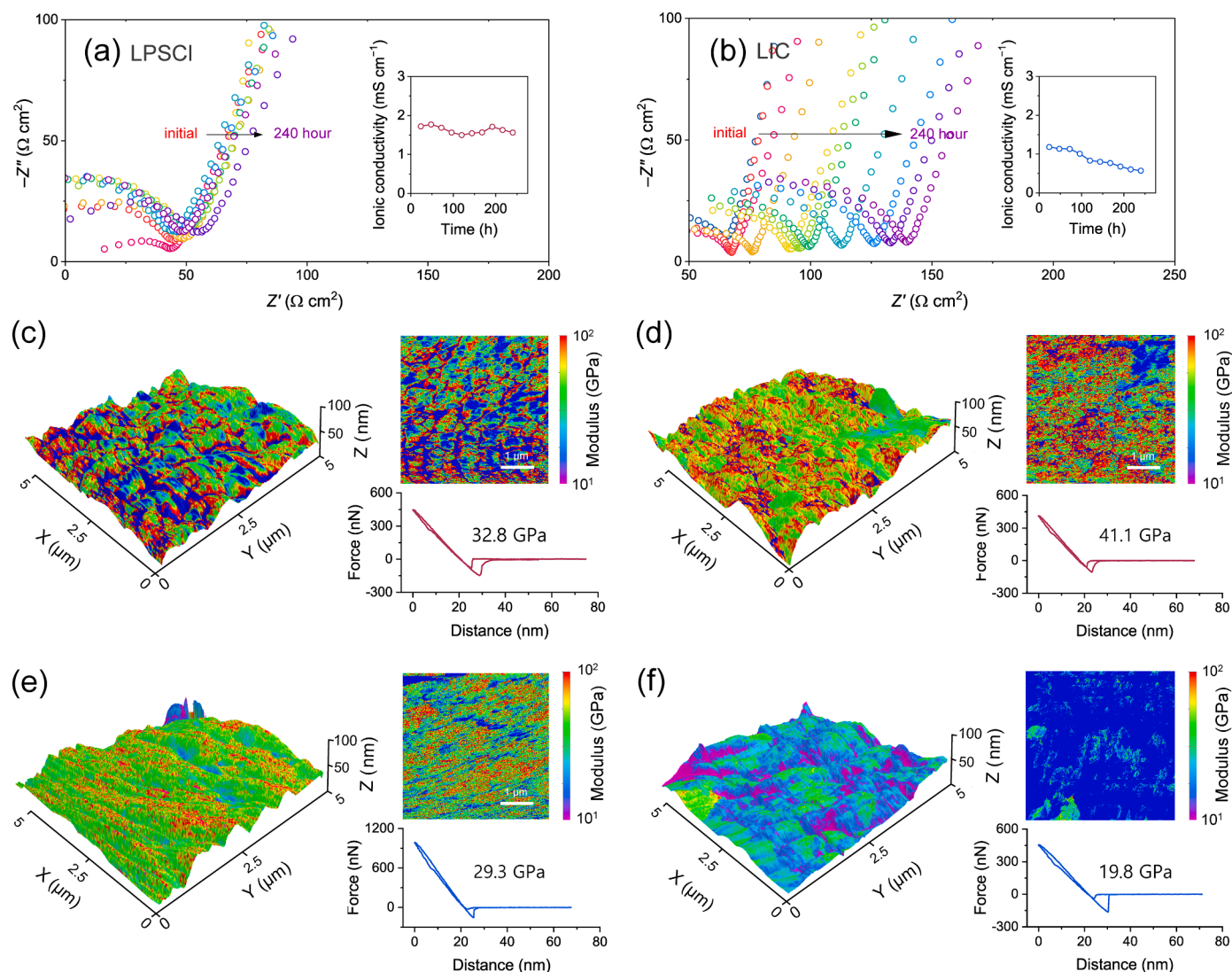


Fig. 4. Evolution of AC-impedance spectra and corresponding ionic conductivities of (a) LPSCI- and (b) LIC-SEs upon aging test. Modulus mapping results of (c, d) pristine and (e, f) tested (c, e) LPSCI and (d, f) LIC.

conductivity at 1.6 mS cm^{-1} without any considerable decay under pressure (Fig. 4b). The following might be considered as possible origins for the conductivity loss of LIC: (i) structure/phase transformation, (ii) compositional change, and (iii) deteriorated solid-solid contacts. We confirmed that LIC retained its crystal structure/phase (Fig. S8) and compositions (Fig. 3d) after continuous conductivity measurements or charge-discharge cycling, from which possibilities (i) and (ii) mentioned above could be ruled out. Interestingly, no conductivity loss was observed in the LIC material that was stored in an Ar-filled glove box for a similar period (Fig. S9). This suggests that the degradation of Li^+ -conducting properties occurred only under pressurized conditions. Thus, it may be closely associated with the mechanical characteristics of the LIC.

The mechanical properties of the LIC were investigated using AFM-based FS, which enabled a quantitative evaluation of the Young's modulus (E), adhesion force, and deformation at a given force [51]. In this study, the E value was calculated from the slope of the force-distance curve obtained on the surface of the SE pellet [52]. The FS measurements were carried out over an area of $5 \mu\text{m} \times 5 \mu\text{m}$, and the resulting distributions of E before and after the conductivity measurements are displayed from Figs. 4c–f. LIC had a mean modulus value (E_{mean}) of $\sim 41.1 \text{ GPa}$ (Fig. 4c), which was higher than that of LPSCI ($\sim 32.8 \text{ GPa}$) (Fig. 4e). However, it underwent severe mechanical degradation under

pressure, i.e., cracking, as revealed by the large decrease in E_{mean} to $\sim 19.8 \text{ GPa}$ (Fig. 4d). This experimental finding was further supported by the SEM images showing the microscale cracks throughout the LIC taken from the pressurized cell (Fig. S10). For comparison, the E_{mean} value of LPSCI remained almost similar ($\sim 29.3 \text{ GPa}$) under pressurized conditions (Fig. 4f). The GPa-level E_{mean} values obtained by the FS measurements are higher than the values that have been usually acquired by practical macroscopic measurements. When a large force is applied to a bulk material during macroscopic tests, the plastic deformation occurs, resulting in a decrease in apparent modulus values. Microscopic techniques, such as AFM-based nanoindentation used in our study, allow for accurate measurements of the intrinsic mechanical properties of the material, since the applied force and test area are significantly smaller than those for macroscopic tests. In fact, the E_{mean} values are consistent with the previous simulation results by Kim et al. [53].

To elucidate the distinct mechanical deforming behavior observed by the FS, we conducted a computational finite element analysis on the formation and propagation of cracks in composite cathodes. When the mechanical behaviors of both LPSCI and LIC were assumed to be fully elastic, NCM-LPSCI and NCM-LIC showed the similar cracking patterns characterized by the onset point (degree of lithiation) for crack initiation and the total crack length (Fig. S11). Given the viscoelastic and plasticity behavior of LPSCI [54], we conducted simulations in consideration of

elastic and/or plastic deformation: (i) elastic-plastic deformation of LPSCI in NCM-LPSCI and (ii) elastic-only deformation of LIC in NCM-LIC (Fig. S12). Fig. 5a shows the von Mises stress distribution in NCM-LPSCI after full lithiation, where the plasticity of LPSCI allows for stress redistribution over a larger region with low stress values (less than 0.2 GPa at the NCM/LPSCI interface). In contrast, in the composite cathode with LIC, which is assumed to deform only elastically, the stress cannot be redistributed and thus accumulates at the NCM/LIC interface with high stress values over 0.5 GPa (Fig. 5b). Consequently, LPSCI had fewer cracks than LIC in the composite cathode as the plastic deformation-assisted stress redistribution reduced the interfacial traction force on the crack surfaces inside the SE material, thereby decreasing the driving force for the formation and propagation of cracks. From the combined experimental and computational studies (impedance decoupling, SSRM, FS, and finite element analysis), we suggest that

LIC effectively stabilizes the interface with NCM but is vulnerable to the cracking caused by the repeated volume change of NCM under pressurized condition. Consequently, ionic transport pathways are gradually disconnected, resulting in a considerable increase in R_{ion} (Fig. 5c). In addition to the electrochemical stability window, therefore, plasticity should be considered to improve the mechanical stability of halide electrolytes. Specifically, future research can explore the use of additives (dopants) and/or modification of the halide structure to ensure intimate solid-solid contacts under pressure while maintaining its high electrochemical stability.

4. Conclusion

We demonstrated the critical degradation factors of halide- and sulfide-SE-based cathodes by impedance decoupling combined with

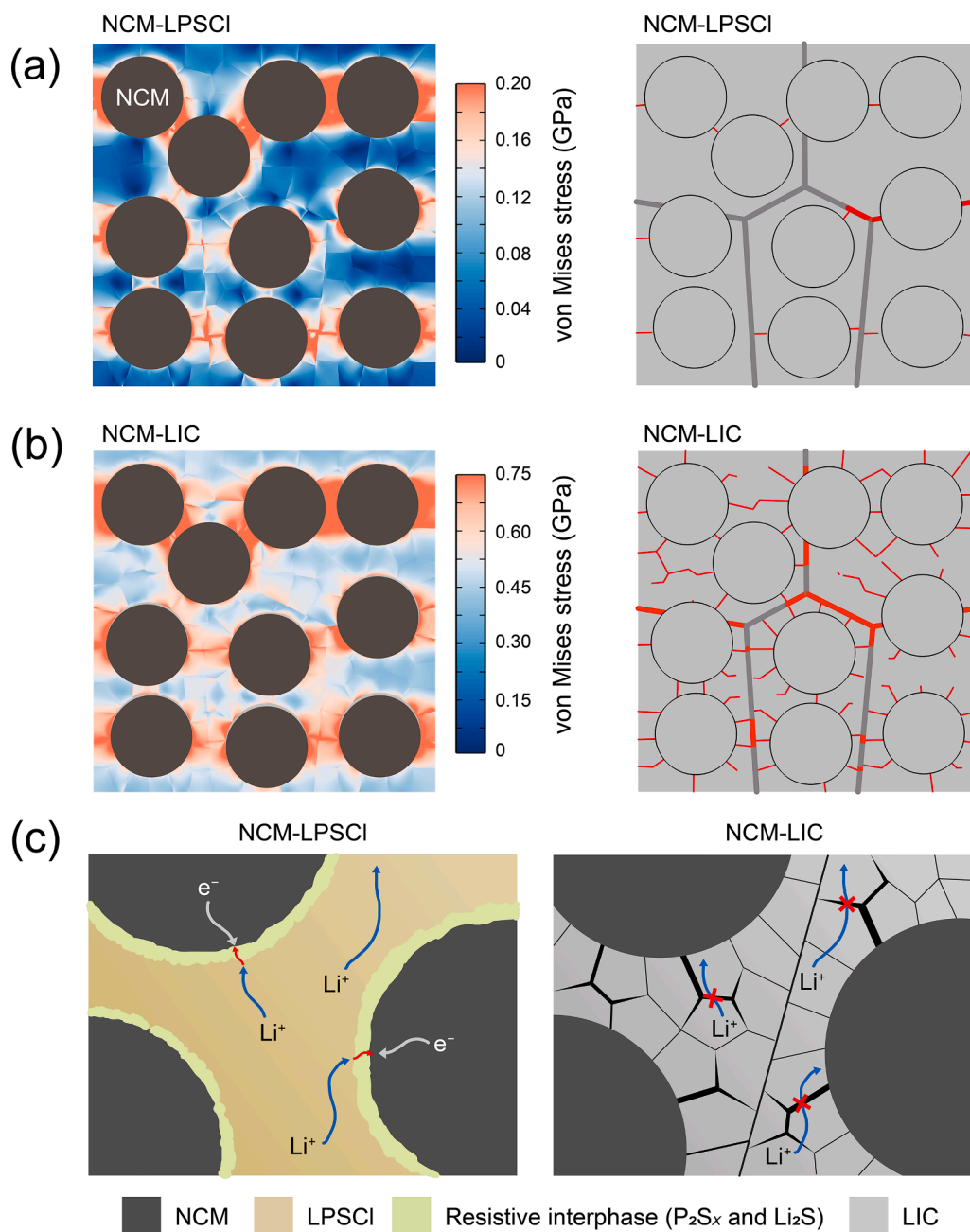


Fig. 5. Distributions of von Mises stress and cracks in the electrolyte phase of (a) NCM-LPSCI and (b) NCM-LIC after full lithiation. The red lines represent the cracks at the intergranular interfaces. (c) Schematic illustration of SE-dependent degradation behaviors of NCM-LPSCI and NCM-LIC.

SSRM, FS, and computational finite element analyses. TLM-based impedance analyses showed significant increases in Li^+ -transport and interfacial impedances for NCM-LIC and NCM-LPSCI, respectively, upon cycling. From the analyses of the electrical and mechanical properties of composite cathodes, we revealed that the degradation of NCM-LIC was mainly driven by the loss of solid-solid contact due to the mechanical deformation of LIC under pressure, while NCM-LPSCI suffered from the formation of resistive interphases. Furthermore, the computational finite element analysis suggested that the deformation behavior of SEs plays a vital role in the formation and propagation of cracks in the composite cathode and that the elastic deformation of LIC would make NCM-LIC vulnerable to cracking caused by the volume change of NCM, leading to the loss of Li^+ -transport pathways. This study provides insights into the complex electrochemo-mechanical degradation in composite cathodes.

CRediT authorship contribution statement

Jonghyeok Yun: Investigation, Writing – original draft. **Hong Rim Shin:** Investigation, Writing – original draft. **Trung Dinh Hoang:** Formal analysis. **Siwon Kim:** Formal analysis. **Jae Hyuk Choi:** Investigation. **Beomsu Kim:** Investigation. **Hyuck Jung:** Investigation. **Janghyuk Moon:** Conceptualization, Writing – review & editing. **Jong-Won Lee:** Conceptualization, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ensm.2023.102787](https://doi.org/10.1016/j.ensm.2023.102787).

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